



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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09.07.2026



Outline

- Executive Summary
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- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Introduction

- Project background and context
 - SpaceX reuses the Falcon 9 first stage instead of discarding it, cutting launch cost
 - Predicting landing success from launch conditions lets us estimate launch cost in advance
- Problems you want to find answers
 - How do payload mass, launch site, and orbit affect landing success?
 - Has the landing success rate improved over time?
 - Which classification algorithm predicts landing outcome most accurately?

Section 1

Methodology

Methodology

Data collection methodology:

- SpaceX REST API calls + Wikipedia web scraping with BeautifulSoup

Perform data wrangling

- Converted raw Outcome text (e.g. “True ASDS”) into a binary Class label

Perform exploratory data analysis (EDA) using visualization and SQL

Perform interactive visual analytics using Folium and Plotly Dash

Perform predictive analysis using classification models

- Logistic Regression, SVM, Decision Tree, KNN - tuned via GridSearchCV, 5-fold CV

Data Collection

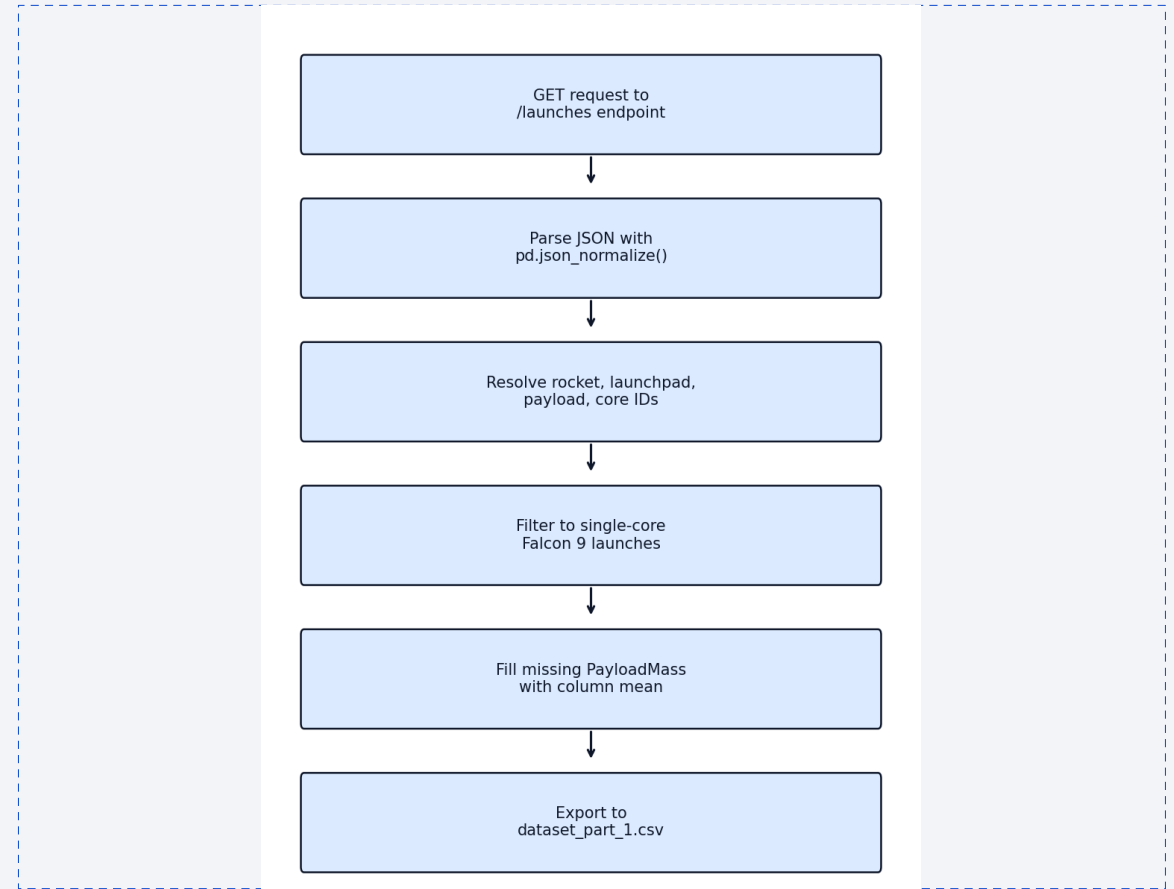
Launch data was collected from two independent sources and cross-referenced:

the SpaceX REST API (structured JSON) and the Wikipedia list of Falcon 9 launches (HTML table, via web scraping).

Together they provided 90 Falcon 9 launches (2010-2020) with booster, payload, site, and outcome details.

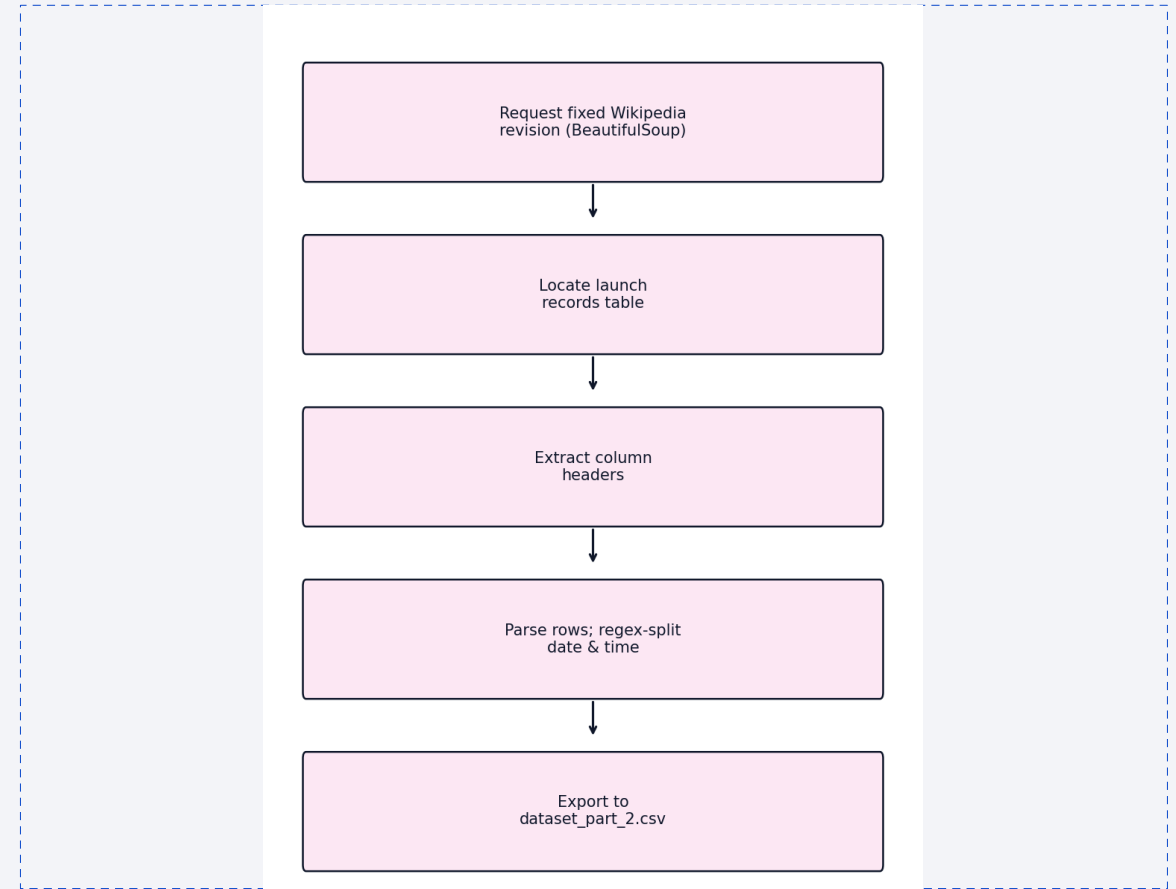
Data Collection – SpaceX API

- GET request to /launches endpoint → `pd.json_normalize()` → resolve rocket/launchpad/payload/core IDs via follow-up calls → filter to single-core Falcon 9 → fill missing PayloadMass with column mean → export to CSV
- GitHub URL:
github.com/ankitarchoudhary/BM-Data-Science-Capstone/blob/main/Phase1_Data_Collection.ipynb



Data Collection - Scraping

- Request a fixed, archived Wikipedia revision (for reproducibility) → locate launch table with BeautifulSoup → extract column headers → parse each row with regex to reliably split combined date-time text → export to CSV



- GitHub URL:
github.com/ankitarchoud

Data Wrangling

The raw Outcome column (e.g. “True ASDS”, “False Ocean”) combines landing success and landing type as text.

Outcomes were mapped to Class=1 (True ASDS / True RTLS / True Ocean) or Class=0 (all False outcomes, None None, None ASDS).

GitHub URL: github.com/ankitarchoudhary/IBM-Data-Science-Capstone/blob/main/Phase2_Data_Wrangling.ipynb

EDA with Data Visualization

Charts plotted: Flight Number vs. Site, Payload vs. Site, Success Rate by Orbit (bar), Flight Number vs. Orbit, Payload vs. Orbit, and Yearly Success Trend (line).

Used to identify which features (site, orbit, payload, time) are most associated with landing success, before modeling.

GitHub URL: github.com/ankitarchoudhary/IBM-Data-Science-Capstone/blob/main/Phase3_EDA_Visualization.ipynb

EDA with SQL

10 SQLite queries were run, covering:

- Unique launch sites; records starting with 'CCA'

- Total & average payload mass, by orbit

- Successful landings with payload 4000-6000 kg

- Total success/failure counts; max-payload boosters

- Unsuccessful 2020 landings; outcome ranking by date range

GitHub URL: github.com/ankitarchoudhary/IBM-Data-Science-Capstone/blob/main/Phase4_EDA_SQL.ipynb

Build an Interactive Map with Folium

Map objects added: Circle markers for each launch site, a MarkerCluster of individual launches color-coded green (success) / red (failure), and a PolyLine with distance label from KSC LC-39A to the nearest coastline.

These let a viewer see site locations, compare outcome density per site, and confirm sites are built near the coast (for safety) at a glance.

GitHub URL:

github.com/ankitarchoudhary/IBM-Data-Science-Capstone/blob/main/Phase5_Interactive_Visual_Analytics.ipynb

Build a Dashboard with Plotly Dash

Dashboard components: a launch-site dropdown, a pie chart of landing outcomes, a payload-mass range slider, and a scatter chart of payload vs. outcome, all interactively linked via callbacks.

Added so a user can filter to any site or payload range and immediately see how success rate and payload relate, without re-running code.

GitHub URL: github.com/ankitarchoudhary/IBM-Data-Science-Capstone/blob/main/Phase5_Interactive_Visual_Analytics.ipynb

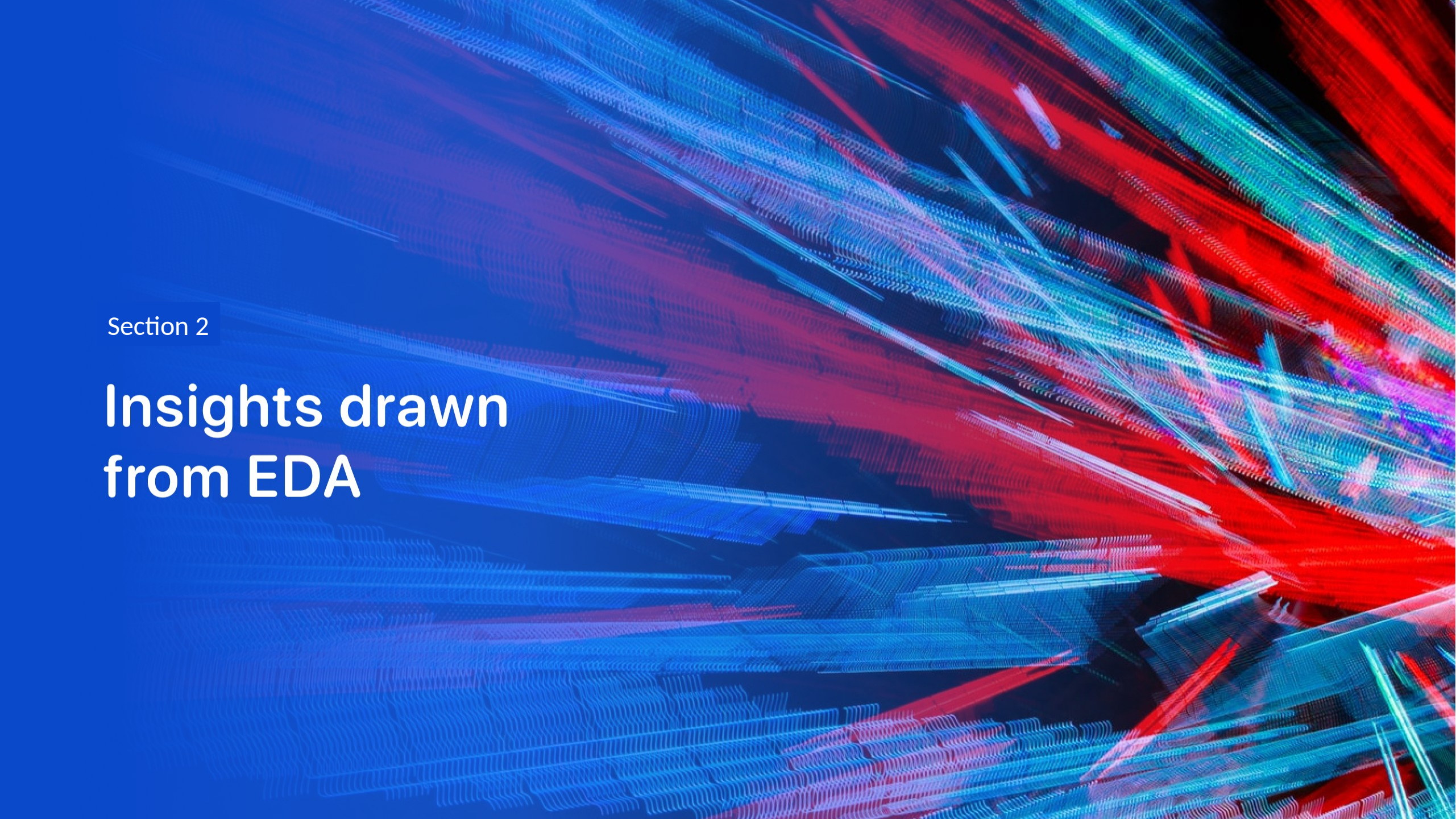
Predictive Analysis (Classification)

Standardized features (one-hot encoded Orbit/LaunchSite) → 80/20 train-test split → GridSearchCV (5-fold CV) tuned Logistic Regression, SVM, Decision Tree, and KNN → compared test accuracy → selected KNN as best (94.4%).

GitHub URL: github.com/ankitarchoudhary/IBM-Data-Science-Capstone/blob/main/Phase6_Predictive_Analysis.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

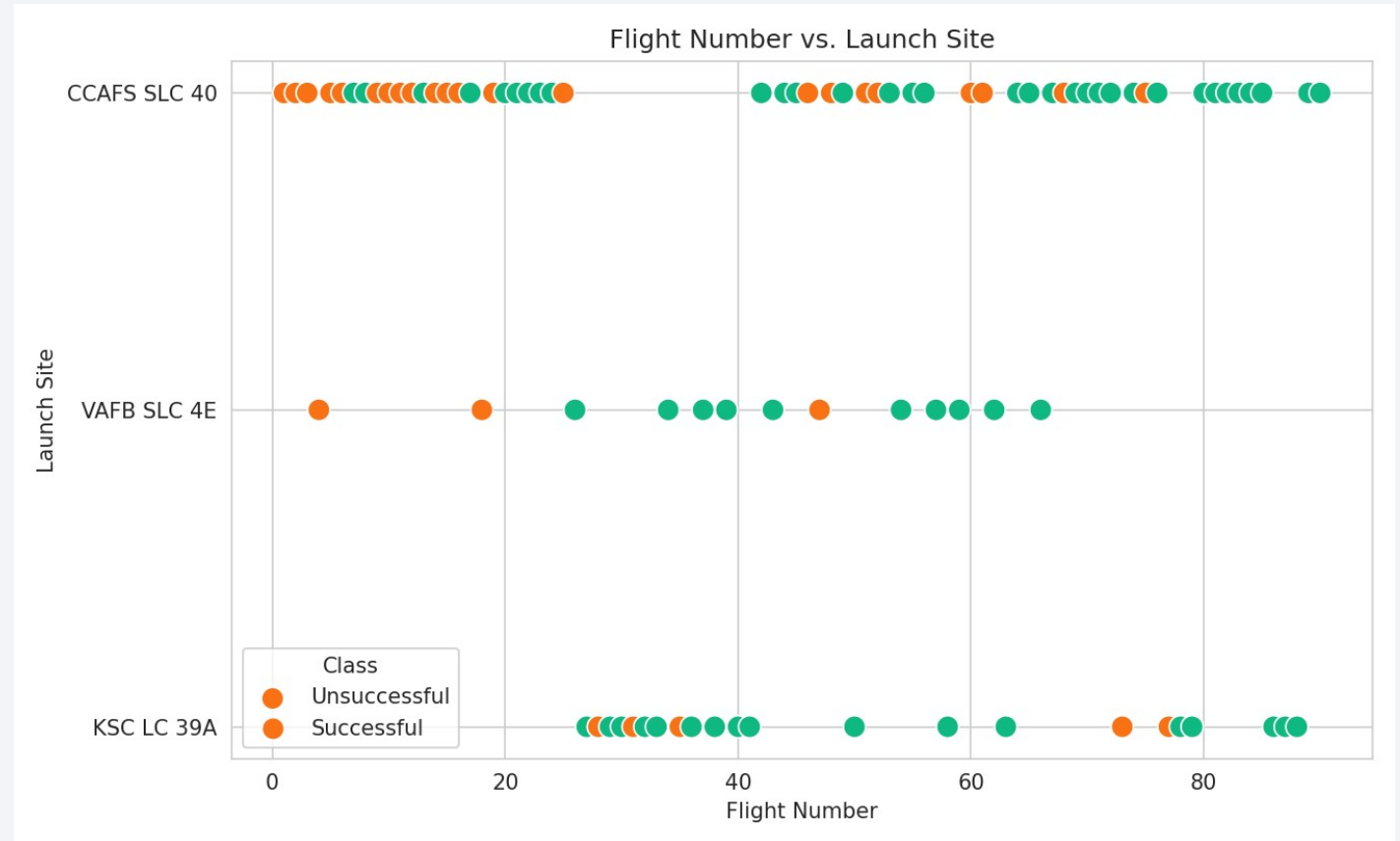
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

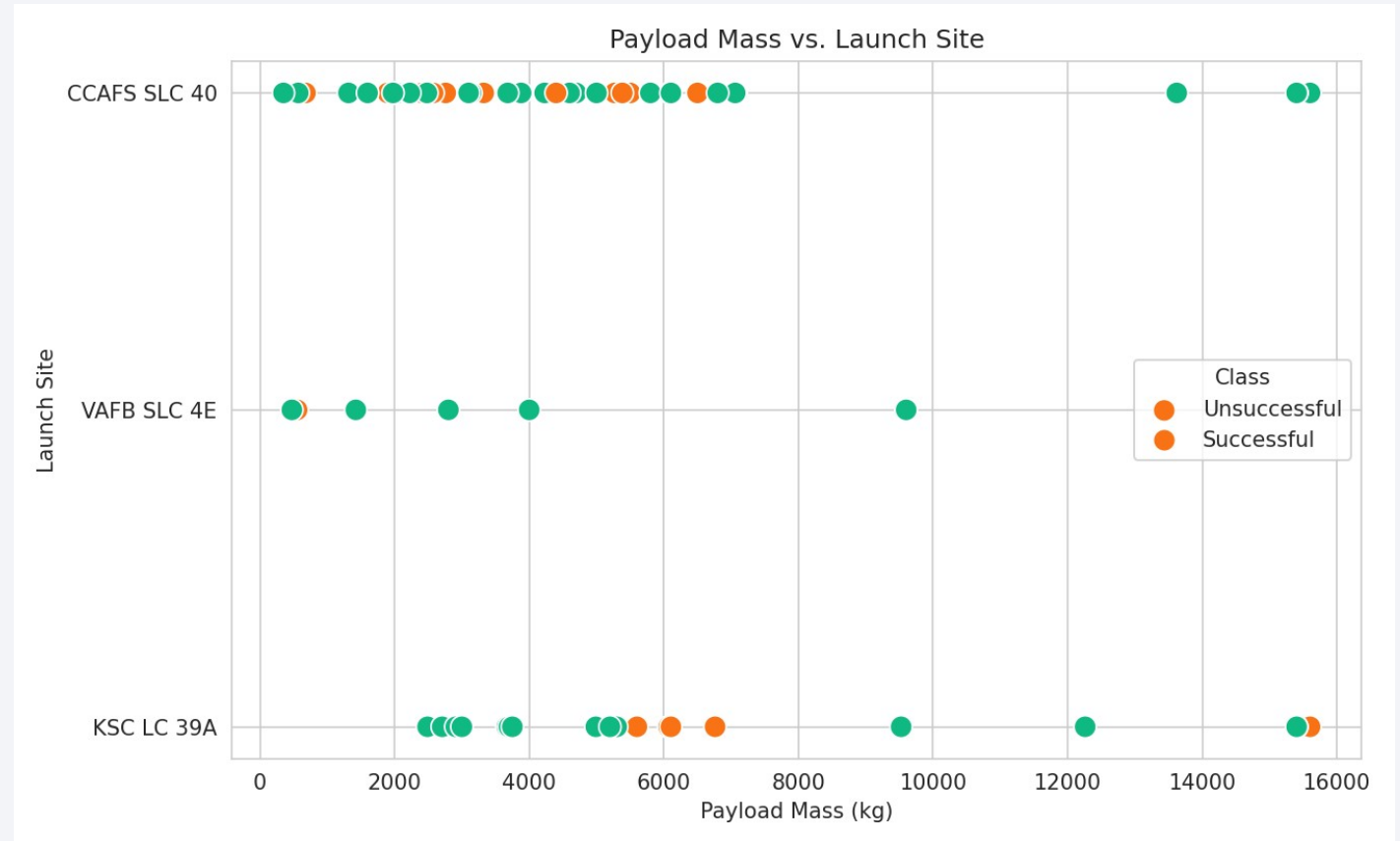
Flight Number vs. Launch Site

- Early flight numbers (1-20) are mostly unsuccessful landings across all sites;
- later flight numbers are dominated by successful outcomes.
- CCAFS SLC-40 has the largest share of launches across the full range.



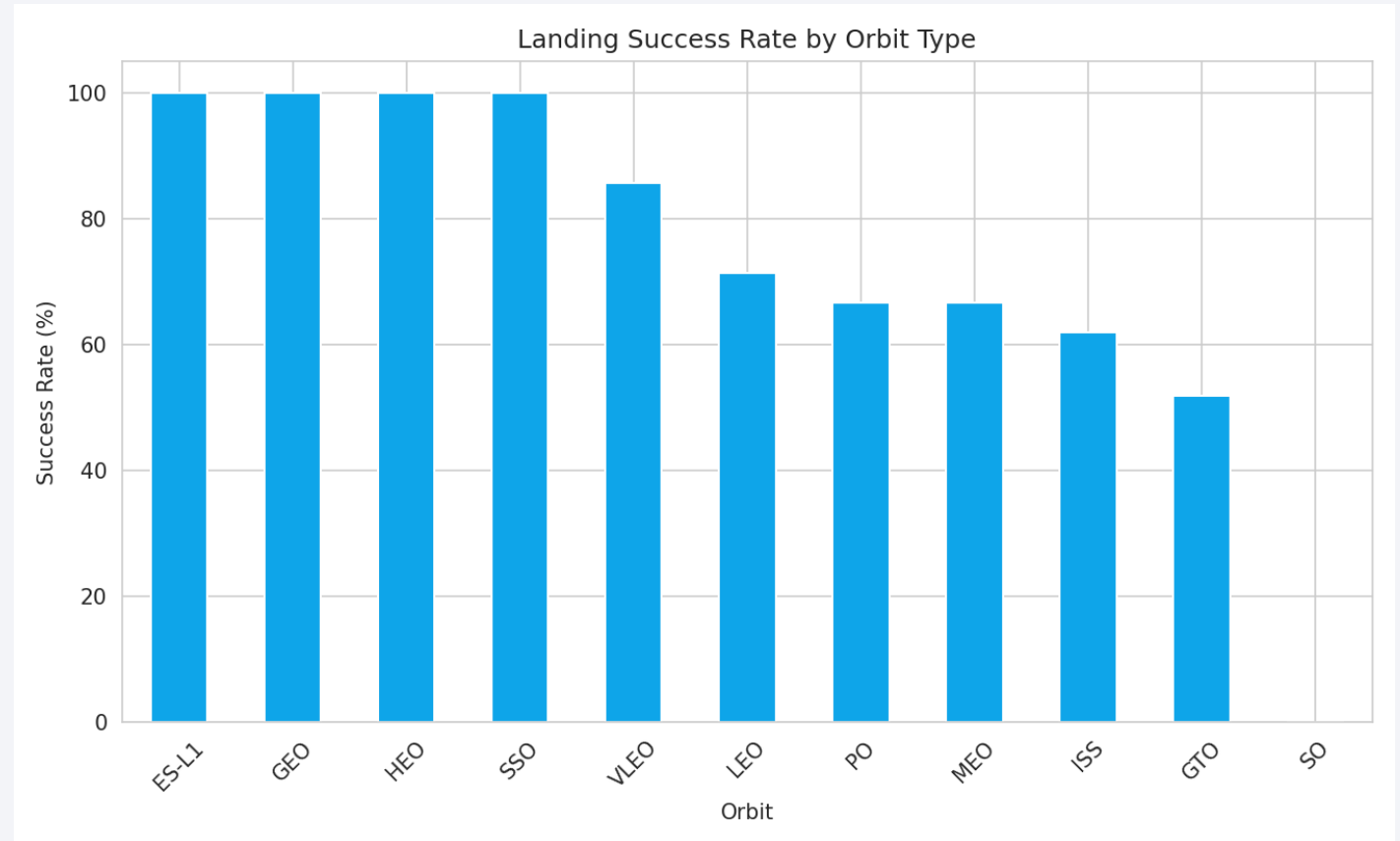
Payload vs. Launch Site

- VAFB SLC-4E and CCAFS SLC-40 handle a wide payload range (under 1000 kg to 15,000+ kg).
- KSC LC-39A tends to handle heavier payloads with a higher share of successes.



Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, SSO orbits: 100% success rate (small sample sizes, 1-5 launches each).
- GTO: most common orbit (27 launches) but lowest success rate at 51.9%.



Flight Number vs. Orbit Type

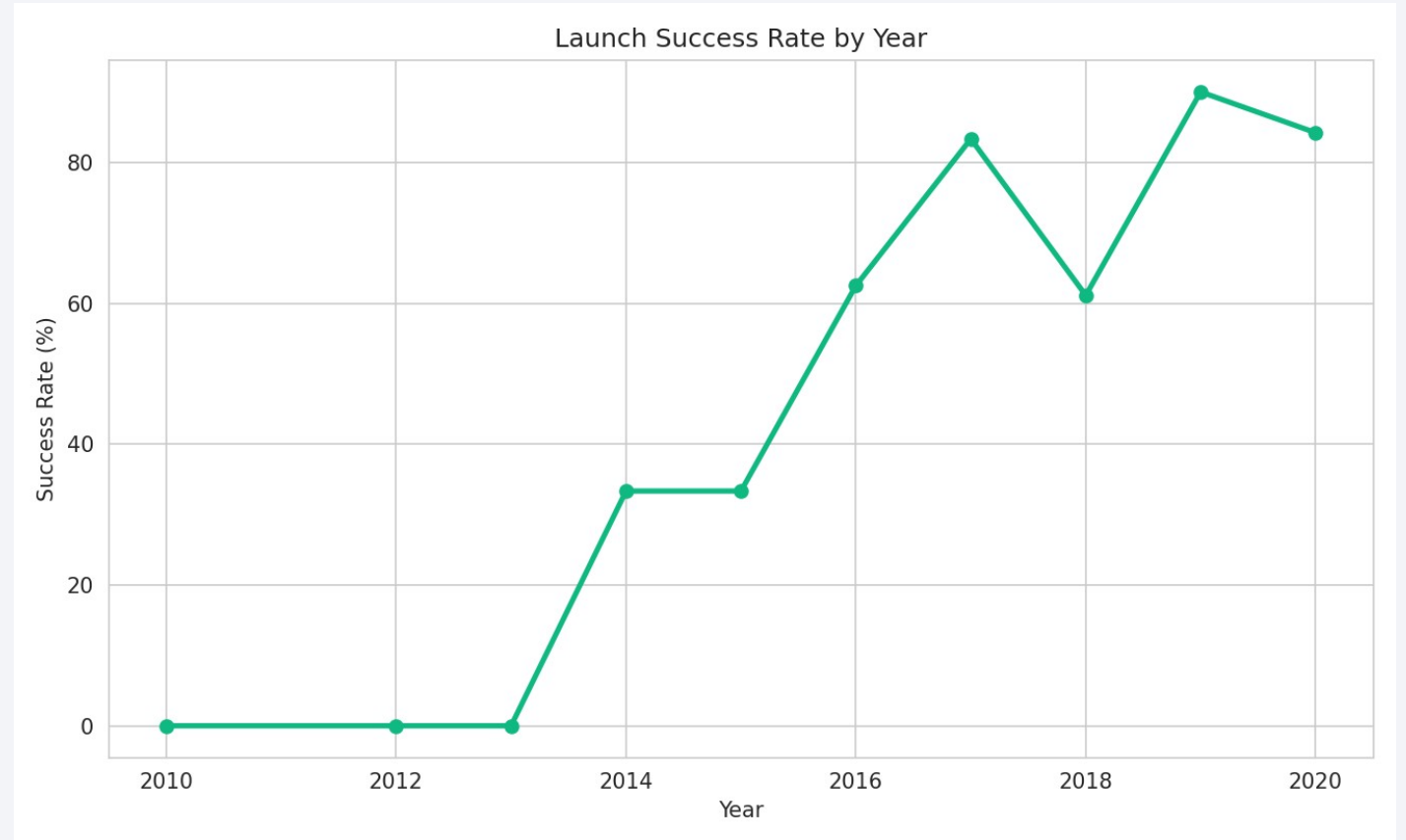
- In the LEO orbit, success appears related to the number of flights (later flights more successful).
- There is no clear relationship between flight number and success when in GTO orbit.

Payload vs. Orbit Type

- Heavy payloads have a negative influence on GTO orbit success.
- Heavy payloads have a positive influence on GTO and Polar LEO (ISS) orbit success.

Launch Success Yearly Trend

- Success rate was 0% from 2010-2013, climbed through 2014-2016, reached 83.3% in 2017
- and peaked at 90% in 2019, confirming SpaceX's improving landing technology over time.



All Launch Site Names

`SELECT DISTINCT LaunchSite FROM spacex;`

Result: 3 unique launch sites — CCAFS SLC 40, VAFB SLC 4E, KSC LC 39A.

Launch Site Names Begin with 'CCA'

```
SELECT * FROM spacex WHERE LaunchSite LIKE 'CCA%' LIMIT 5;
```

Result: 5 records returned, all from CCAFS LC-40 / CCAFS SLC-40, dated 2010-2013, with payload mass 0-677 kg.

Total Payload Mass

```
SELECT SUM(PayloadMass) FROM spacex;
```

Result: total payload mass across all 90 launches = 549,446.3 kg.

Average Payload Mass by F9 v1.1

`SELECT Orbit, AVG(PayloadMass) FROM spacex GROUP BY Orbit;`

Result: overall average payload mass across all launches = 6,105.0 kg (varies by orbit; VLEO and GTO orbits carry the heaviest average payloads).

First Successful Ground Landing Date

```
SELECT MIN(Date) FROM spacex WHERE Class = 1;
```

Result: first successful landing occurred on 2014-04-18.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT Serial, PayloadMass FROM spacex WHERE Class=1 AND  
PayloadMass BETWEEN 4000 AND 6000;
```

Result: several boosters landed successfully with payload in this range, confirming mid-weight payloads are compatible with successful recovery.

Total Number of Successful and Failure Mission Outcomes

```
SELECT Class, COUNT(*) FROM spacex GROUP BY Class;
```

Result: 60 successful landings, 30 unsuccessful — an overall 66.7% success rate.

Boosters Carried Maximum Payload

```
SELECT Serial FROM spacex WHERE PayloadMass = (SELECT  
MAX(PayloadMass) FROM spacex);
```

Result: boosters B1048, B1049, and B1051 each carried the maximum payload mass of 15,600 kg.

2015 Launch Records

SELECT Date, Serial, LaunchSite, Outcome FROM spacex WHERE
Class=0 AND Date LIKE '2020%';

Result: 3 unsuccessful landings recorded in 2020 (Jan, Feb, Mar), at
KSC LC-39A and CCAFS SLC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Outcome, COUNT(*) ... WHERE Date BETWEEN '2015-01-01'  
AND '2018-01-01' GROUP BY Outcome ORDER BY COUNT(*) DESC;
```

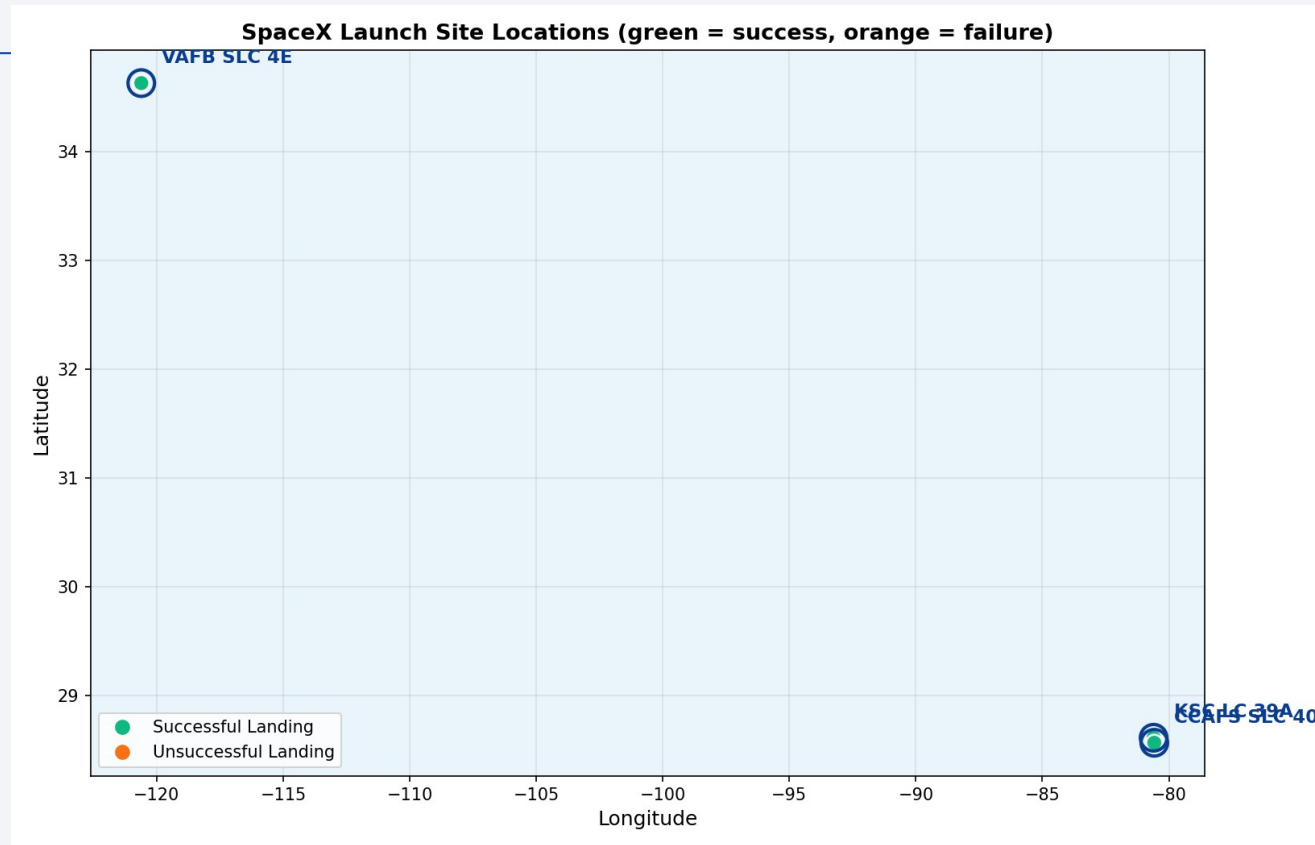
Result: True ASDS (12) was the most common outcome in this window, followed by True RTLS (8).

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

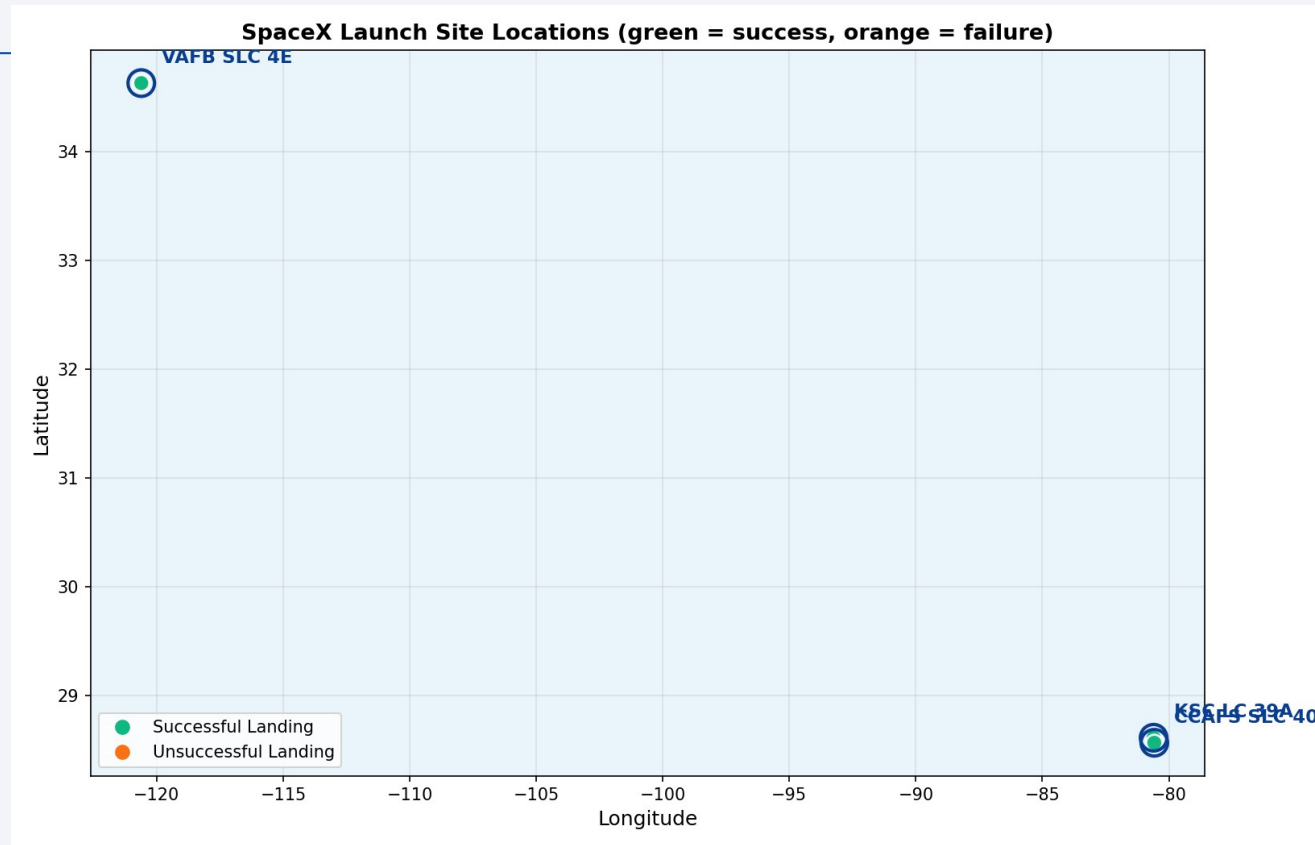
All Launch Sites - Location Markers



All 3 launch sites plotted with circle markers and text labels at their real coordinates.

CCAFS SLC-40 / KSC LC-39A sit on Florida's east coast; VAFB SLC-4E sits on California's west coast.

Color-Labeled Launch Outcomes



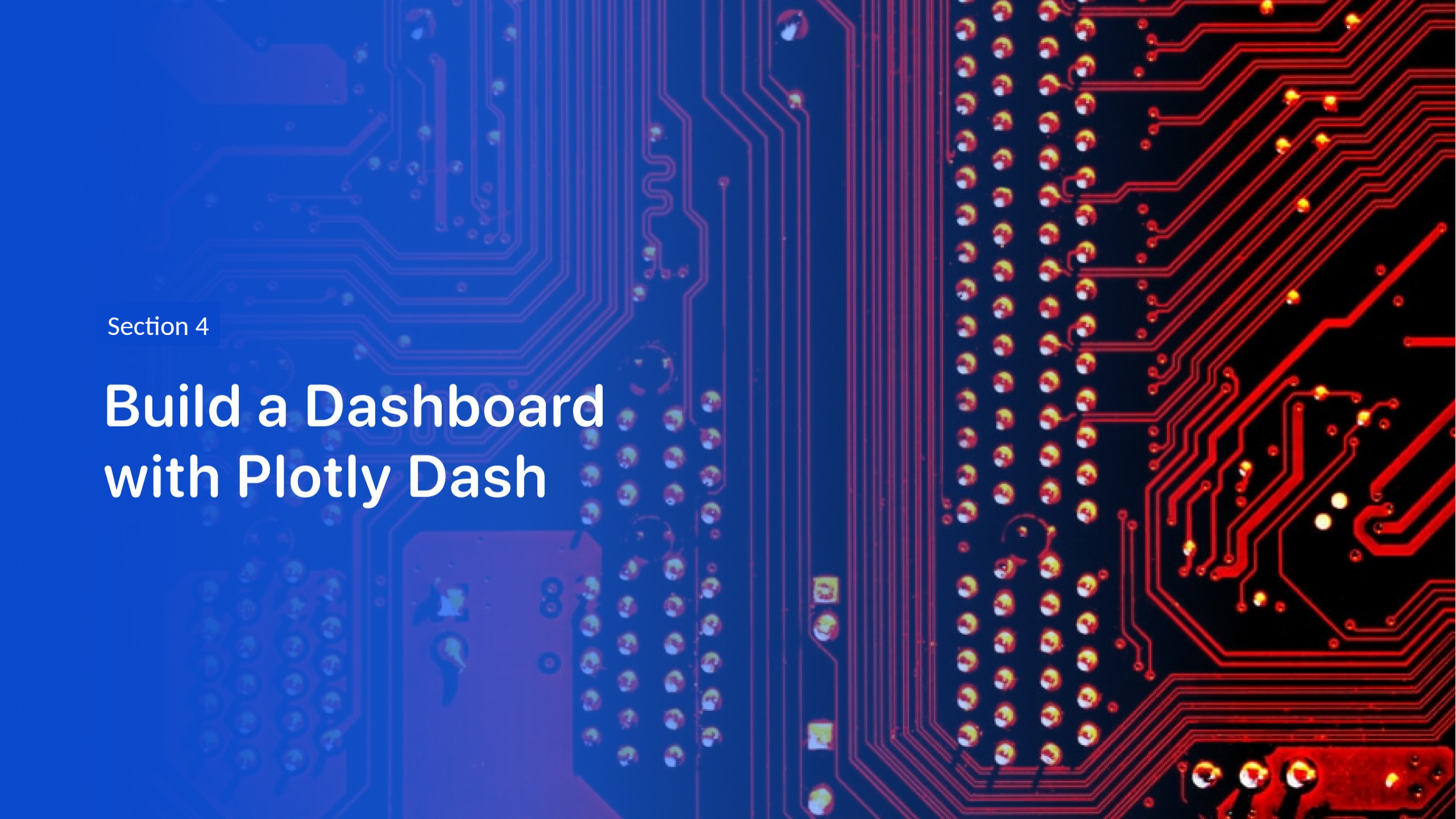
Markers color-coded green (successful landing) and red (unsuccessful), clustered per site.

KSC LC-39A shows a visibly higher share of green markers, matching its 77.3% success rate.

Proximity Analysis - KSC LC-39A to Coastline

Distance from KSC LC-39A to the nearest coastline point calculated at 5.86 km using the Haversine great-circle formula.

All launch sites sit close to the coast, so a failed rocket falls over open water rather than a populated area.



Section 4

Build a Dashboard with Plotly Dash

Total Successful Landings by Site (Pie Chart)



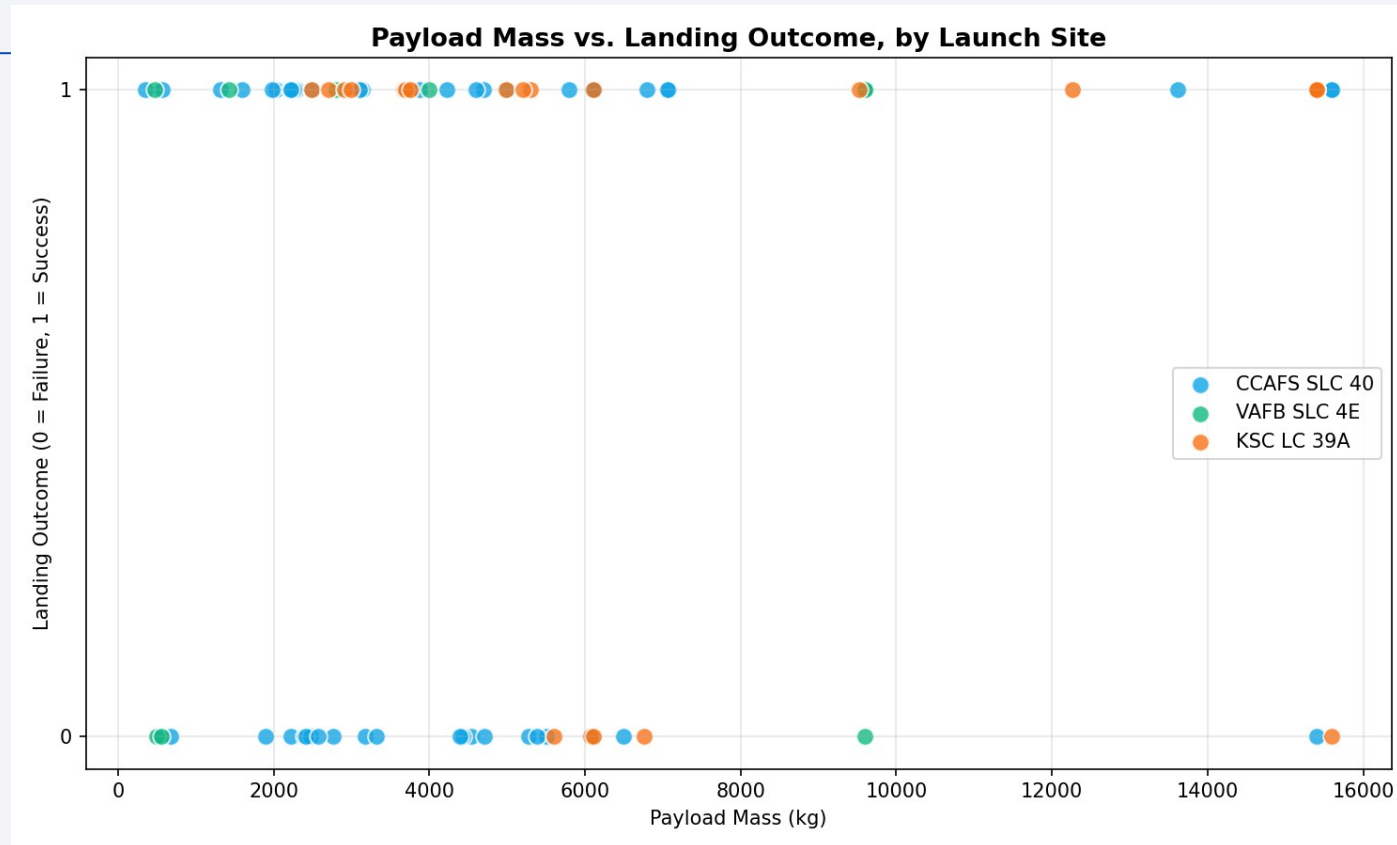
CCAFS SLC-40 contributes the largest share of successful landings (33 of 60), simply because it has the most total launches.

Highest Success-Ratio Site - KSC LC-39A



KSC LC-39A has the highest launch-site success ratio in the dataset at 77.3% (17 of 22 launches).

Payload Mass vs. Launch Outcome (Scatter, All Sites)



Successes and failures appear across most of the payload range; VAFB SLC-4E launches cluster more tightly around successful outcomes.

The interactive range slider lets a user filter this view to a specific payload window. 41

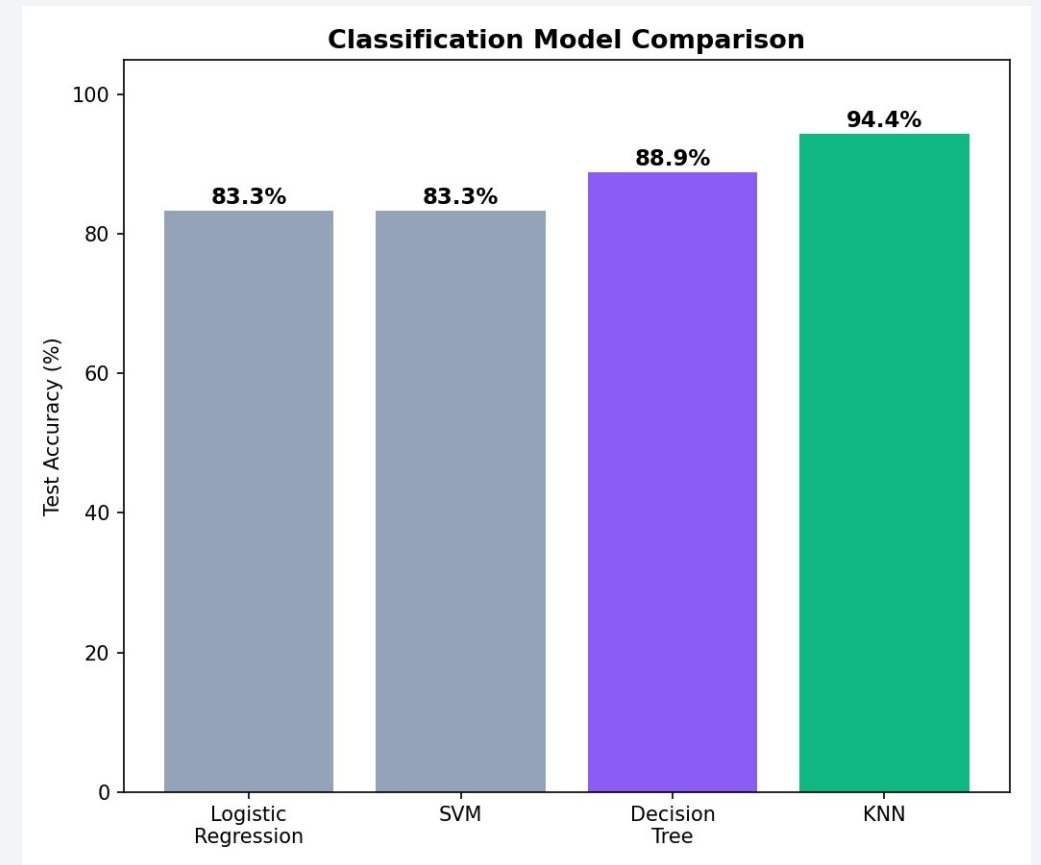


Section 5

Predictive Analysis (Classification)

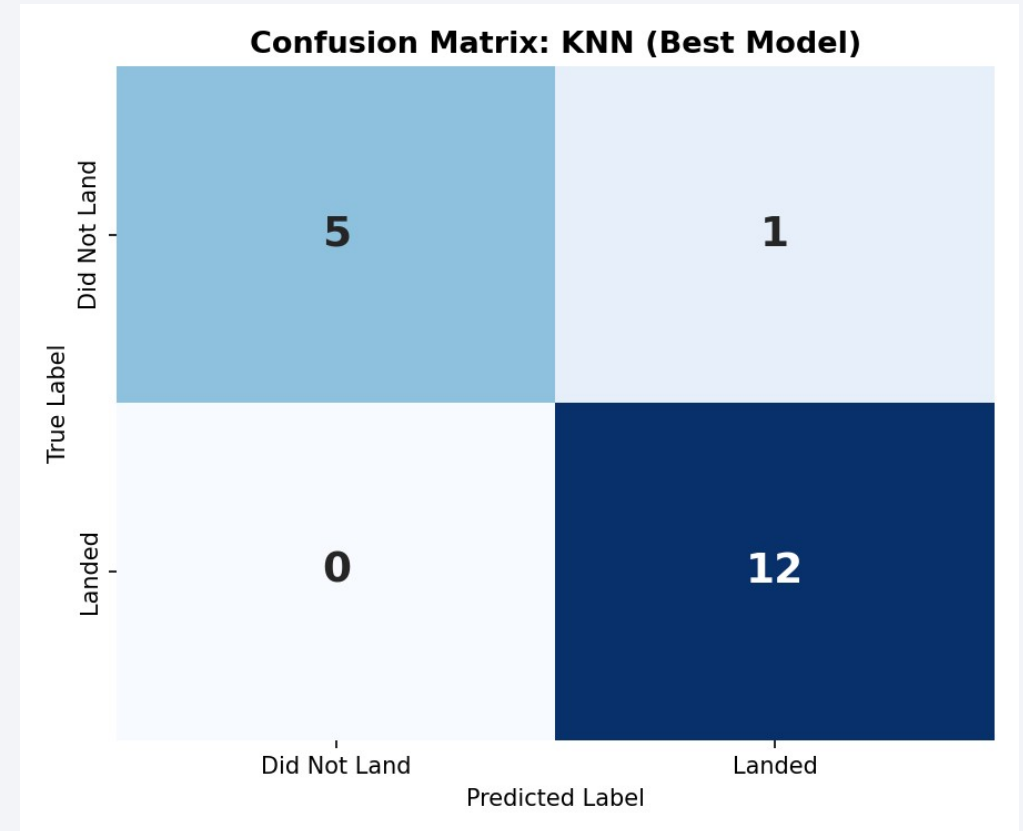
Classification Accuracy

- 4 models tuned with GridSearchCV (5-fold CV): Logistic Regression, SVM, Decision Tree, KNN.
- KNN achieved the highest test accuracy at 94.4% (n_neighbors=7, Manhattan distance).



Confusion Matrix

- KNN (best model) classified 17 of 18 test launches correctly (94.4% accuracy).
- Zero false negatives: every actual successful landing was correctly identified.
- Only 1 false positive: a launch predicted to land that did not.



Conclusions

KNN predicts Falcon 9 first-stage landing success with 94.4% test accuracy and zero false negatives.

KSC LC-39A is the most reliable launch site (77.3% success); CCAFS SLC-40 is used most often but has the lowest rate (60.0%).

GTO, the most common orbit, has the lowest success rate (51.9%) - future cost reductions likely depend on improving GTO-specific landing profiles.

Landing success rate rose from 0% (2010-2013) to 84-90% (2019-2020), a visible timeline of SpaceX's engineering learning curve.

SQL, visualization, and ML analysis all independently point to the same conclusions on site, orbit, and time, increasing confidence in the results.

Appendix

Full GitHub repository (all notebooks, datasets, and code):
github.com/ankitarchoudhary/IBM-Data-Science-Capstone

Thank you!

